



## EE214 Digital Circuits Laboratory

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Problem set: 3

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# Fibonacci Number Detector

## Instructions

1. Use structural modelling for this experiment i.e. instantiate components and use port map to connect those components.
2. For the design part do pen-paper design with kmap and get it verified by your TA/RA.
3. In pen paper design use proper labeling for each wire. And use the same labels for the VHDL code.
4. Perform RTL simulation using the provided testbench and tracefile.
5. Perform manual verification using Max-10 board.
6. Demonstrate the experiment to your TA/RA.
7. Submit the entire project files in .zip format in Moodle.

## Design

A Fibonacci Number Detector identifies whether a given binary number belongs to the Fibonacci sequence. Such detectors are useful in digital systems for sequence recognition, hardware acceleration, and pattern matching applications where specific numerical values must be identified efficiently.

Design and implement a 5-bit Fibonacci Number Detector. The circuit should output '1' only when the 5-bit binary input represents a Fibonacci number between 0 and 31.

Show the pen-paper design and minimized expression using K-Maps to the corresponding evaluating TAs.

## VHDL Description

Write a VHDL description for the given problem statement.

Inputs(5-bit):  $x_4x_3x_2x_1x_0$

Output(1-bit):  $y$

Tracefile format:  $\langle x_4x_3x_2x_1x_0 \rangle \langle y \rangle 1$

## Design Verification

- Perform RTL Level Simulation of the Majority Circuit using the generic testbench to confirm the correctness of your description using the [TRACEFILE](#)
- Pin plan switches S5 to S1 as input and LED 8 as output and show the correctness of the design on board.

## Testing on Board

- Test the correctness of your design on the Xen-10 board.
- Perform the pin mapping as follows:

$$\begin{array}{lcl} x_4-x_0 & \implies & SW5-SW1; \\ y & \implies & LED8 \end{array}$$